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GBA-Related Biomarkers in Parkinson's disease

Analysis of plasma sphingolipids: ceramides, glucosylceramides, lactosylceramides and sphingomyelins

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PPMI Project ID: 135

Rev Oct 5 2011





Summary (or Abstract)

Mutations in the β -glucocerebrosidase gene (GBA) modulates cellular accumulation of potentially pathogenic glycosphingolipids and progression of Parkinson's disease. We have quantified sphingolipids related to the glucocerebrosidase pathway by liquid chromatography-tandem mass spectrometry (LC-MS/MS), namely ceramides (Cer), glucosylceramides (GlcCer), lactosylceramides (GL2) and sphingomyelins (SM) in human plasma.

Method

Lipid Extraction from Human Plasma

Sphingolipids in plasma were extracted by protein precipitation in extraction solution (acetonitrile/methanol/formic acid, 50/50/0.2, v/v/v, with 5 mM ammonium formate) using Hamilton STAR liquid handler (Reno, NV). The extraction solvent contained 1 ng/mL of C12:0 Cer, 1.34 ng/mL of C12:0 GlcCer, and 1.7 ng/mL of C12:0 GL2 as internal standards (IS) cocktails (Sphingolipid Mix I, Avanti polar lipids, Alabaster, AL). The mixtures were vortexed and centrifuged. The resulting supernatants were transferred to 384-well plates for Cer, GlcCer, and GL2 analysis. Twenty μ L of the resulting supernatant was also mixed with 1 mL of mobile phase A (Acetonitrile/Methanol/Acetic acid/Water, 96/2/1/1, v/v/v/v, with 5 mM ammonium Acetate) for SM analysis using ThermoMultidrop Combi (Thermo Fisher Scientific, Grand Island, NY) and Apricot I-pipette (Covina, CA). Mobile phase A contained 5 ng/mL of C12:0 SM (Avanti polar lipids, Alabaster, AL) as an IS.

Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS) Analysis Ceramides (Cer), glucosylceramides (GlcCer), and lactosylceramides (GL2)

Quantitative analysis of sphingolipid was performed by using liquid chromatography with tandem mass spectrometry (LC-MS/MS). Cer, GlcCer, and GL2 were analyzed using a Waters 2.1 x 100 mm Cortecs HILIC column at a flow rate of 0.5 mL/min and a Waters Acquity binary solvent system (Waters Corporation, Milford, MA). Mobile phase A and mobile phase B (79/20/1, methanol/water/acetic acid, v/v/v, with 5 mM ammonium acetate) were used. The LC eluents were analyzed by a triple quadrupole mass spectrometer (AB Sciex API 5000, Foster City, CA) in MRM mode. Calibration curves were generated with C16:0 Cer, C18:0 Cer, C20:0 Cer, C22:0 Cer, C24:1 Cer, C24:0 Cer, C16:0 GL2, C18:0 GL2, C24:1 GL2, and C24:0 GL2 (Avanti polar lipids, Alabaster, AL) and GlcCer standards (Matreya, LLC, State College, PA).

Sphingomyelins (SM)

SM was analyzed using a Waters 2.1 x 100 mm Cortecs HILIC column at a flow rate of 0.5 mL/min and a Waters Acquity binary solvent system (Waters Corporation, Milford, MA). Mobile phase A and Mobile phase B were used to separate SM from phosphatidylcholines (PC). The LC eluents were analyzed by a triple quadrupole mass spectrometer (AB Sciex API 4000, Foster City, CA) in MRM mode. Calibration curves were generated with C16:0 SM, C18:0 SM, C24:1 SM, and C24:0 SM standards (Avanti polar lipids, Alabaster, AL) and C20:0 SM and C22:0 SM standards (Matreya, LLC, State College, PA).





For each sphingolipid seven species are reported (C16, C18, C20, C22, C23, C24:1 and C24). The sum of these species is reported as Total sphingolipid (i.e. Total Cer, Total GlcCer, Total GL2 and Total SM).

TM LLOQ

(ug/ml in human plasma)

Ceramide	C16 Cer	C18 Cer	C20 Cer	C22 Cer	C23 Cer	C24:1 Cer	C24 Cer
	0.04	0.01	0.01	0.01	0.01	0.02	0.01

Glucosylceramide	C16 GlcCer	C18 GlcCer	C20 GlcCer	C22 GlcCer	C23 GlcCer	C24:1 GlcCer	C24 GlcCer
	0.01	0.01	0.01	0.02	0.01	0.01	0.04

Lactosylceramide	C16 GL2	C18 GL2	C20 GL2	C22 GL2	C23 GL2	C24:1 GL2	C24 GL2
	0.06	0.03	0.06	0.03	0.03	0.03	0.03

SM LLOQ

(ug/ml in human plasma)

Sphingomyelin	C16 SM	C18 SM	C20 SM	C22 SM	C23 SM	C24:1 SM	C24 SM
	1.95	0.50	1.00	0.50	0.50	1.95	0.50

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GBA-Related Biomarkers in Parkinson's disease

Analysis of cerebrospinal fluid sphingolipids: glucosylceramides, lactosylceramides and sphingomyelins

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Summary (or Abstract)

Mutations in the β -glucocerebrosidase gene (GBA) modulates cellular accumulation of potentially pathogenic glycosphingolipids and progression of Parkinson's disease. We have quantified sphingolipids related to the glucocerebrosidase pathway by liquid chromatography-tandem mass spectrometry (LC-MS/MS), namely glucosylceramides (GlcCer), lactosylceramides (GL2) and sphingomyelins (SM) in human cerebrospinal fluid (CSF).

Method

Lipid Extraction from Human CSF

GlcCer and GL2 in CSF were extracted by liquid-liquid extraction. Briefly, 50 μ L of human CSF were suspended in 1 mL methanol, 1.5 mL chloroform, and 2 mL water. Methanol contained 0.25 ng/mL of C12:0 GlcCer, and 1.25 ng/mL of C12:0 GL2 as internal standards (IS) (Avanti polar lipids, Alabaster, AL). After vortexing and centrifugation, the lower phase was carefully transferred, dried with N₂ gas and resuspended in 0.2 mL of solvent (Acetonitrile/Methanol, 90/10, v/v) for GlcCer and GL2 analysis.

For SM analysis, 10 μ L of CSF were extracted by protein precipitation in extraction solution (Acetonitrile/Methanol/Acetic acid/Water, 96/2/1/1, v/v/v/v, with 5 mM ammonium Acetate) using Hamilton STAR liquid handler (Reno, NV). The extraction solvent contained 5 ng/mL of C12:0 SM (Avanti polar lipids, Alabaster, AL) as an IS. The mixtures were vortexed and centrifuged. The resulting supernatants were transferred to 384-well plates for SM analysis.

Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS) Analysis of

Glucosylceramides (GlcCer) and Lactosylceramides (GL2)

Quantitative analysis of sphingolipids was performed using LC-MS/MS. GlcCer, and GL2 were analyzed using a Waters 2.1 x 100 mm Cortecs HILIC column at a flow rate of 0.5 mL/min and a Waters Acquity binary solvent system (Waters Corporation, Milford, MA). Mobile phase A consisted of Acetonitrile/Methanol/Acetic acid/Water = 96/2/1/1, v/v/v/v, and 5 mM ammonium acetate. Mobile phase B consisted of methanol/Water/Acetic acid = 79/20/1, v/v/v, and 5 mM ammonium acetate. GlcCer was also separated from galactosylceramide (GalCer). The LC eluents were analyzed by a triple quadrupole mass spectrometer (AB Sciex API 5000, Foster City, CA) in MRM mode. Calibration curves were generated with GL2 of C16:0, C18:0, C24:1, and C24:0 (Avanti polar lipids, Alabaster, AL) and GlcCer standards (Matreya, LLC, State College, PA).





LC-MS/MS Analysis of Sphingomyelins (SM)

SM was analyzed using a Waters 2.1 x 100 mm Cortecs HILIC column at a flow rate of 0.5 mL/min and a Waters Acquity binary solvent system (Waters Corporation, Milford, MA). Mobile phase A and mobile phase B were used to separate SM from phosphatidylcholines (PC). The LC eluents were analyzed by a triple quadrupole mass spectrometer (AB Sciex API 4000, Foster City, CA) in MRM mode. Calibration curves were generated with SM of C16:0, C18:0, C24:1, and C24:0 (Avanti polar lipids, Alabaster, AL) and C20:0 and C22:0 (Matreya, LLC, State College, PA).

For each sphingolipid seven species are reported (C16, C18, C20, C22, C23, C24:1 and C24). The sum of these species is reported as total sphingolipid (i.e. Total GlcCer, Total GL2 and Total SM).

LLOQ

(ng/ml in human CSF)							
Glucosylceramide	C16 GlcCer	C18 GlcCer	C20 GlcCer	C22 GlcCer	C23 GlcCer	C24:1 GlcCer	C24 GlcCer
	0.08	0.04	0.20	0.32	0.20	0.44	0.60
Lactosylceramide	C16 GL2	C18 GL2	C20 GL2	C22 GL2	C23 GL2	C24:1 GL2	C24 GL2
	1.56	1.56	1.56	1.56	1.56	1.56	1.56
Sphingomyelin	C16 SM	C18 SM	C20 SM	C22 SM	C23 SM	C24:1 SM	C24 SM
	10	10	10	10	10	10	20





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Analysis of cerebrospinal fluid sphingolipids: Ceramide

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Summary (or Abstract)

Mutations in the β -glucocerebrosidase gene (GBA) modulates cellular accumulation of potentially pathogenic glycosphingolipids and progression of Parkinson's disease. We have quantified ceramides related to the glucocerebrosidase pathway by liquid chromatography-tandem mass spectrometry (LC-MS/MS).

Method

Lipid Extraction from Human CSF

CSF ceramides (Cer) were extracted by protein precipitation. Briefly, 30 μ L of human CSF were suspended in 60 μ L methanol and 30 μ L chloroform. Methanol contained 1 ng/mL of C12:0 Cer as an internal standard (IS) (Avanti polar lipids, Alabaster, AL). After vortexing and centrifugation, the supernatant was dried with N_2 gas and resuspended in 120 μ L of solvent (Acetonitrile/Methanol, 90/10, v/v) for Cer analysis.

Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS) Analysis of

Ceramides (Cer)

Quantitative analysis of Cer was performed using LC-MS/MS. Cer was analyzed using a Waters 2.1 x 100 mm Cortecs HILIC column at a flow rate of 0.5 mL/min and a Waters Acquity binary solvent system (Waters Corporation, Milford, MA). Mobile phase A consisted of Acetonitrile/Methanol/Acetic acid/Water = 96/2/1/1, v/v/v/v, and 5 mM ammonium acetate. Mobile phase B consisted of methanol/Water/Acetic acid = 79/20/1, v/v/v, and 5 mM ammonium acetate. The LC eluents were analyzed by a triple quadrupole mass spectrometer (AB Sciex API 5000, Foster City, CA) in MRM mode. Calibration curves were generated with Cer of C16:0, C18:0, C20:0, C22:0, C24:1, and C24:0 (Avanti polar lipids, Alabaster, AL).

LLOQ

Ceramide	(ng/ml in human CSF)						
	C16 Cer	C18 Cer	C20 Cer	C22 Cer	C23 Cer	C24:1 Cer	C24 Cer
	0.05	0.05	0.05	0.05	0.05	0.05	0.05

Note:

Due to technical difficulties the Ceramide analysis was completed after the teams of Prof. Alcalay (Columbia) and Prof. Scherzer (BWH, Harvard) were unblinded and the data posted to LONI. The unblinding key was not shared with the Sanofi team.

